NiT Sparkathon 2026

Challenge Handbook

10 curated challenges

50 participants | 10 teams | 4 - 5 people per team

Collab. Build your next startup idea. Go viral.

12–13 November 2026 | NiT | TUHH | Hamburg

Supported by Sparkasse Harburg-Buxtehude, KPMG, Lovable AI and Impossible Founders

How the challenges work

Each challenge sets out a problem worth solving and the core idea it is built around. How you solve it, for whom and with which technology is up to your team.

Every brief has the same short structure:

- **Context:** why the problem matters now.
- **The challenge:** what you are asked to build.
- **Core concept:** the idea your solution must embody. This is the only fixed element.
- **Guiding question:** one question to start your thinking, not a checklist to complete.

What every team delivers

These requirements are the same for every challenge, so all tracks are comparable in ambition.

1. **A working prototype.** A complete user journey with a meaningful AI, agent, data-analysis or automation component at its core. A static interface or an AI-generated report alone is not enough.
2. **Evidence that it works.** A small set of test cases you define yourself, including at least one edge or failure case, and a comparison with a simple baseline such as manual work, rules-only logic or a single unassisted model.
3. **Responsible design.** Be transparent about evidence, uncertainty and limitations. Users approve any consequential action before it happens.
4. **A business or impact case.** Who uses it, who pays, what value it creates and what would need to be validated next.
5. **A live test.** The jury may provide an unseen input within the scope you declare. A useful result or a well-justified refusal both count; a scripted happy path does not.

Ground rules

- No dedicated hardware is required. All challenges are software-first.
- Lovable AI must be used for the interface and application workflow. Teams may add their own code, APIs, databases and agent frameworks, and remain responsible for working backend logic.
- Disclose every AI platform, library, dataset and API you use, as the official rules require. Label any mocked integration; it cannot replace the central working capability.
- Use sandboxes for external actions. Never use real payments, sensitive personal data or unapproved communications.
- Comparisons involving people should report how many took part. A small pilot is prototype evidence, not statistical proof.

Team allocation

There are ten teams of five, and each curated challenge should be taken by one team. A team that chooses the DIY route replaces one curated assignment, so one curated challenge then remains unassigned.

Challenge menu

1. The Agent Economy Startup
2. The Consultant's Agent Team
3. The Superpower Arena
4. The Centaur Test
5. Trust, Explained
6. Measurable Impact Engine
7. Trust Between Machines
8. From Lab to Market
9. Engineering the Trade-off
10. Your Agent, Your Side

1. The Agent Economy Startup

CONTEXT

Most software waits for instructions. Agent-based systems can coordinate work, use tools and act across services through protocols such as Agent-to-Agent Communication (A2A) and the Model Context Protocol (MCP). This opens up business models that conventional apps and service companies cannot run economically.

THE CHALLENGE

Build an AI-native startup whose core service runs on autonomous agents, and show it delivering a real outcome for a customer.

CORE CONCEPT

Agents are the business model, not a feature. Remove them and the service should no longer work. Show customer value, a revenue model and market potential.

> **GUIDING QUESTION**
>
> **What becomes a viable business only when agents can do the work?**

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2. The Consultant's Agent Team

CONTEXT

AI can already produce a polished slide deck. What it does not produce on its own is consulting value: a diagnosis grounded in evidence, advice that adapts when the facts change, and a professional who stands behind it.

THE CHALLENGE

Design an agent-supported consulting service that changes how consulting is delivered, not just how quickly deliverables are produced.

CORE CONCEPT

Recommendations stay linked to evidence and evolve as the client's situation changes. A consultant remains accountable for the final judgment.

> **GUIDING QUESTION**
>
> **What consulting service becomes possible in 2030 that clients cannot buy today?**

3. The Superpower Arena

CONTEXT

Claims that AI beats humans, or that humans still win, are easy to make and hard to interpret. The outcome depends on the task, the rules and what each side is allowed to use.

THE CHALLENGE

Build a playable competition that pits humans against AI and reveals where each side has the advantage.

CORE CONCEPT

A fair contest across more than one capability, transparent scoring, and an insight that goes beyond naming a winner. No winner is assumed in advance.

> **GUIDING QUESTION**
>
> **What would a truly fair match between a human and a machine look like?**

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4. The Centaur Test

CONTEXT

Pairing people with AI can improve results, or it can lead people to accept wrong answers with more confidence. Most tools are not designed to tell the difference.

THE CHALLENGE

Build a platform that compares humans alone, AI alone and humans working with AI, and shows when collaboration helps and when it hurts.

CORE CONCEPT

Measure reliance as well as performance: when AI advice is followed, ignored, or followed too readily.

> **GUIDING QUESTION**
>
> **When does AI assistance make people worse?**

5. Trust, Explained

CONTEXT

Deepfakes, generated text and manipulated media are cheap to produce. A simple real or fake label is not enough for a business to act on. People need to see why something should or should not be trusted.

THE CHALLENGE

Build a verification product that assesses content, evaluates its sources and produces a transparent trust score that users can inspect and challenge.

CORE CONCEPT

Every assessment is explainable down to the evidence, honest about uncertainty, and resistant to deliberate manipulation.

> **GUIDING QUESTION**
>
> **What would it take for someone to trust your trust score?**

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6. Measurable Impact Engine

CONTEXT

Many social-impact products count activity, such as users, clicks or downloads, rather than change. Without a credible way to measure impact, good ideas struggle to attract funding or scale.

THE CHALLENGE

Build an AI solution for a specific group facing a societal challenge of your choice, for example in education, healthcare, sustainability, mobility or democratic participation, with impact measurement designed into the product.

CORE CONCEPT

A clear beneficiary, a concrete change and a metric that separates what was observed from what is projected. Safeguards for vulnerable users are part of the design.

> **GUIDING QUESTION**
>
> **How would you prove your solution made a difference?**

7. Trust Between Machines

CONTEXT

As agents begin to negotiate, transact and exchange data with other agents, confident claims and self-reported reputations are not enough. An agent economy needs a way for machines to establish trust with one another.

THE CHALLENGE

Build a trust layer that lets agents decide whom to trust, detects unreliable or deceptive behavior, and responds before damage is done.

CORE CONCEPT

Trust rests on verifiable evidence rather than self-assertion, and decisions are explained to the people the agents work for.

> **GUIDING QUESTION**
>
> **How does an agent earn trust, and how does it lose it?**

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8. From Lab to Market

CONTEXT

Many technical inventions never become products, often because their creators lack the time or support to assess the commercial opportunity. Confident but unsupported market figures can make early decisions worse rather than better.

THE CHALLENGE

Build an agent-supported tool that turns a technical idea into an honest commercial assessment and a plan for validating it.

CORE CONCEPT

Facts, estimates and assumptions are clearly separated, and the weakest assumptions become the next experiments.

> **GUIDING QUESTION**
>
> **What would a founder need to know before investing the next six months in an idea?**

9. Engineering the Trade-off

CONTEXT

Engineering and business decisions balance cost, performance, risk and sustainability, and stakeholders weigh these factors differently. A persuasive AI narrative is no substitute for sound calculations and explicit trade-offs.

THE CHALLENGE

Build a decision-support product that makes trade-offs visible and helps stakeholders reach, and record, better decisions.

CORE CONCEPT

Transparent, reproducible logic underneath; AI for explanation and facilitation on top. Users can see how the preferred option shifts when priorities or assumptions change.

> **GUIDING QUESTION**
>
> **How can a team disagree productively and still decide?**

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10. Your Agent, Your Side

CONTEXT

Individuals regularly deal with institutions that have more resources and expertise than they do, such as insurers, authorities and service providers. Institutions increasingly use AI on their side; most individuals have nothing comparable.

THE CHALLENGE

Build a personal agent that acts on behalf of an individual in a situation of your choice, from understanding their case to preparing a concrete action.

CORE CONCEPT

The user stays in control: nothing is submitted without their approval, and the agent recognizes when a qualified professional must take over.

> **GUIDING QUESTION**
>
> **What would change if everyone had an expert on their side?**

Judging criteria

Official scoring

All challenges, including DIY, are scored on the same five criteria from the official rules:

- **Innovation & Creativity (35%):** How original and bold is the idea? Does it uniquely solve the problem?
- **Technical Execution (20%):** Does the digital prototype work reliably? Is the software implementation skillful and impressive?
- **Design & User Experience (15%):** How does it feel to use? Is the design thoughtful, intuitive and engaging?
- **Presentation (20%):** How well did the team communicate their vision and demonstrate their project?
- **Community Engagement (10%):** The number of likes on the team's official LinkedIn project post by the submission deadline.

Evidence quality, robustness and the live test inform the existing criteria, mainly Technical Execution. They are not an additional scoring category.